

# ANOGS

## Anti-Cheat Runtime — Deep Reverse-Engineering Analysis

Defensive structural analysis of the **anogs** iOS framework (ARM64 Mach-O dynamic library).

*Initialization → Environment Collection → Detection → Integrity → Classification → Event → Dispatch → Report → Queue → Send.*

|                              |                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------|
| Target                       | anogs (@rpath/anogs.framework/anogs)                                                                      |
| Type                         | Mach-O 64-bit ARM64 dynamic library (iOS), 2,871,200 bytes                                                |
| Code ( <code>__text</code> ) | 0x00004190 - 0x0023A318 (~2.31 MB, 579,682 instructions)                                                  |
| Vendor lineage               | Tencent TSS / ACE (PluginTssSDKLifecycle; <a href="https://down.qq.com/iedsafe">down.qq.com/iedsafe</a> ) |
| Recovered functions          | 10,551 total / ~3,028 reachable anti-cheat core                                                           |
| Method                       | llvm-objdump + capstone/lief; custom function-recovery, call-graph & xref engine                          |
| Scope                        | Read-only, evidence-based. No patching / bypass / hooking performed or described.                         |

## Contents

| #  | Section                                                    |
|----|------------------------------------------------------------|
| 1  | Executive Summary                                          |
| 2  | Architecture                                               |
| 3  | Initialization                                             |
| 4  | Detection Modules                                          |
| 5  | Integrity System (CRC / memory / opcode / code)            |
| 6  | Anti-Instrumentation (debugger / hook / dylib / jailbreak) |
| 7  | MRPCS Subsystem                                            |
| 8  | Event System (IDs & dispatching)                           |
| 9  | Reporting Pipeline                                         |
| 10 | Network / Send Pipeline                                    |
| 11 | Threads                                                    |
| 12 | Function Database                                          |
| 13 | Critical Functions                                         |
| 14 | Full Call Graph                                            |
| 15 | Event-Flow Examples                                        |
| 16 | Evidence Appendix                                          |
| ★  | Anti-Cheat Master Map                                      |

**Confidence legend.** Each finding is tagged: **Confirmed** (structure + strings + data-flow all agree) · **High** (strong disassembly evidence, one small inference) · **Medium** (consistent but partly inferred) · **Unknown**. Stripped binary: only 19 symbols survive, so unnamed routines use Ghidra-style `sub_<offset>` names; proposed names are analyst labels.

# 1 Executive Summary

**anogs** is the native runtime of a mobile anti-cheat product (Tencent TSS / ACE family — confirmed by the Objective-C class `PluginTssSDKLifecycle`, the update endpoint `http://down.qq.com/iedsafe/Client/ios` recovered by decrypting an XOR-0x18 string, and the `MRPCS_IOS` platform tag). It ships as a stripped ARM64 iOS dynamic library that a host game links against and drives through a small C ABI (the `AnoSDK*` exports).

The runtime is organised as a classic collect → detect → classify → report pipeline built around an internal message/event subsystem the vendor calls **MRPCS**. The most important single routine is `sub_102d48` — a 79-case jump-table classifier that maps an integer *event id* to a short tag and a human description and emits it. From that one function the entire event vocabulary was recovered (CRC integrity failures, inline-hook / opcode-mismatch, signature match failures, rule-execution failures, worker-thread start failures, mmap/send-pipeline faults).

## Capabilities present (evidenced by imported syscalls + confirmed routines):

| Capability                | Primary evidence                                                                                  | Location                           | Conf.     |
|---------------------------|---------------------------------------------------------------------------------------------------|------------------------------------|-----------|
| Anti-debug                | <code>sysctl(KERN_PROC, KERN_PROC_PID, getpid) P_TRACED</code> self-check                         | <code>sub_37004</code>             | Confirmed |
| Process monitor           | <code>sysctl(KERN_PROC_ALL)</code> full process-table enumeration                                 | <code>sub_9d29c</code>             | Confirmed |
| Injected-module scan      | <code>_dyld_get_image_name/header</code> loop over loaded images                                  | <code>sub_16f54c/16faf0</code>     | Confirmed |
| Injected-code scan        | <code>mach_vm_region_recurse + proc_regionfilename</code> region→file map                         | <code>sub_2034ec/1f32b4</code>     | Confirmed |
| Substrate / jailbreak     | decrypted <code>cy:com.saurik.substrate;</code> <code>access/stat/lstat/statfs</code> path checks | <code>sub_13a724/13a42c</code>     | High      |
| Inline-hook / opcode      | event ids 10-11 <code>set_inline_hook_error / inline_hook_opcode_dismatch</code>                  | <code>evt 0x0a/0x0b</code>         | Confirmed |
| ObjC swizzle detect       | <code>class_getInstanceMethod + method IMP compare</code>                                         | <code>sub_1eb5a0</code>            | High      |
| Code/data integrity (CRC) | event id 0 <code>ms_data_crc / mrpcs_data_crc_error</code> ; crc report line                      | <code>evt 0x00 / sub_156f8</code>  | Confirmed |
| Downloadable rules        | <code>rule_exe_fail / rule_op_is_change</code> ; MRPCS download thread + hash notify              | <code>evt 0x45/0x4d</code>         | Confirmed |
| Keychain device ID        | <code>SecItemAdd/CopyMatching + iDev fingerprint + /dev/random</code>                             | <code>sub_33d6c</code>             | High      |
| Network reporting         | native <code>socket/connect/send/rcv</code> + host <code>tss_sdk_send_data_to_svr</code> callback | <code>sub_f8210 / sub_f0590</code> | Confirmed |

**Two report exfiltration paths.** Classified events are serialised and delivered either (a) directly over a native socket (`sub_f8210 send / sub_f829c rcv`), or (b) handed back to the host game through the `AnoSDKGetReportData / AnoSDKDelReportData` queue and the registered `tss_sdk_send_data_to_svr` callback. Server responses re-enter through `AnoSDKOnRecvData`, which is how new detection *rules* are delivered.

**False-positive note (per the analysis rules).** The strings "Start Of Scan", "Invalid progressive parameters", "Sampling factors too large" belong to a statically-linked `libjpeg`, not to anti-cheat scanning; they were explicitly excluded. Presence of a string was never treated as proof of a routine's purpose — every functional claim is anchored to instructions, xrefs or data-flow.

## 2 Architecture

The library exposes a flat C ABI (19 AnoSDK\* exports) plus one Objective-C lifecycle class. Internally it is a layered pipeline. The extracted structure (not assumed — derived from call edges and the event classifier) is:

| Layer                      | Responsibility (evidenced)                                                                                                                                                          | Key offsets                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| A. Host ABI                | AnoSDKInit / InitEx / SetUserInfo / OnPause / OnResume / Ioctl / OnRecvData / GetReportData / DelReportData                                                                         | 19 exports @ 0xf0fd0+                           |
| B. Lifecycle bridge (ObjC) | PluginTssSDKLifecycle + CloudAppLifecycleObserver hook applicationDidBecomeActive / DidEnterBackground / WillEnterForeground => drive OnPause/OnResume; DoCmdOnMainThread:/SendCmd: | __objc_data                                     |
| C. Core singletons         | get_context (pthread_once) sub_6bbfc [204 callers]; manager accessor sub_18068 [170]; log/format sub_f0490 [73]                                                                     | sub_6bbfc / sub_18068                           |
| D. Collectors              | device fingerprint (sysctl hw.machine/Model, keychain), process list, dyld image list, VM region map                                                                                | sub_9d29c / sub_16f54c / sub_2034ec / sub_33d6c |
| E. Detectors               | debugger (P_TRACED), inline-hook/opcode, ObjC swizzle, substrate/jailbreak paths, downloadable rule engine                                                                          | sub_37004 / sub_1eb5a0 / ...                    |
| F. Integrity               | CRC/hash of data & code sections; opcode comparison; mmap/vm_copy code mirror                                                                                                       | evt 0x00 / sub_156f8 / sub_1488b8               |
| G. MRPCS event core        | get_event_id sub_11ac6c -> classify+emit sub_102d48 (79-case) -> format sub_102b84 (name=.. desc=..) -> router sub_234b4                                                            | sub_102d48                                      |
| H. Dispatch                | event_dispatch_loop sub_11af80 (self-iterating; blr vtable dispatch)                                                                                                                | sub_11af80                                      |
| I. Report queue            | GetReportData sub_1a25c / DelReportData sub_1a0cc / OnRecvData sub_1a278                                                                                                            | sub_1a25c                                       |
| J. Senders                 | native socket send sub_f8210 / recv sub_f829c / connect sub_f8148; host callback tss_sdk_send_data_to_svr sub_f0590                                                                 | sub_f8210 / sub_f0590                           |

Data flows top-to-bottom; server rules flow back up via layer J→I→E. The MRPCS core (G) is the hinge: collectors/detectors raise conditions, MRPCS assigns each an *event id*, classifies it to (tag, description), and feeds the router/queue/senders.

### 3 Initialization

The library registers **59 constructors** in `__mod_init_func` (0x284b78) that run at load, before any host call. The first is `sub_1013c`; others seed the C++ singletons, RNG and the MRPCS tables. Host-driven initialisation then enters through the exports:

| Export                                     | Offset  | Internal target                                                                            | Notes                                         |
|--------------------------------------------|---------|--------------------------------------------------------------------------------------------|-----------------------------------------------|
| <code>_AnoSDKInit</code>                   | 0xf0fd0 | <code>sub_2d69c</code> (real init) → tail <code>sub_376a0</code> (guard/return marshaller) | <code>w0=int cfg</code> ; returns success bit |
| <code>_AnoSDKInitEx</code>                 | 0xf0ffc | tail <code>sub_2d6cc</code>                                                                | extended init                                 |
| <code>_AnoSDKSetUserInfo</code>            | 0xf1000 | <code>sub_24770</code> ; checks <code>(uid&lt;0x64) &amp;&amp; (info!=0)</code>            | registers user/role; W0 validated             |
| <code>_AnoSDKSetUserInfoWithLicense</code> | 0xf104c | <code>sub_2d92c</code>                                                                     | init with license blob                        |
| <code>_AnoSDKRegistInfoListener</code>     | 0xf118c | <code>sub_260ec</code>                                                                     | installs info callback (uses new/delete)      |

`sub_2d69c` (the real init) calls `sub_25ad8` (subsystem bootstrap; uses `getpid+kill` — a liveness/anti-tamper guard), then the global-context accessors `sub_6bbfc/sub_18068`. Both are `pthread_once` singletons and are the two most-referenced functions in the binary (204 / 170 callers) — i.e. the shared runtime context.

#### Evidence — `AnoSDKInit` entry (0xf0fd0)

```

f0fd0: stp x20,x19,[sp,#-0x20]!      ; prologue
f0fdc: mov x19, x0                   ; save cfg arg
f0fe0: bl 0x2d69c                    ; -> real init (sub_2d69c)
f0fe4: mvn w8, w19                   ; derive return
f0fe8: lsr w0, w8, #31               ; W0 = success bit
f0ff8: b 0x376a0                     ; tail: shared guard/return

```

*Confidence: High. Public entry is a thin wrapper over `sub_2d69c`; W0 is a boolean success returned to the host.*

## 4 Detection Modules

Detectors were identified by clustering functions on their imported-syscall fingerprint and then confirming each by disassembly (not by strings). Each row is a distinct detector; “raises” names the MRPCS event it ultimately feeds where that link is established.

| Detector               | Location                                          | Mechanism (disassembly-verified)                                                      | Conf.     |
|------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------|-----------|
| Debugger (P_TRACED)    | sub_37004                                         | sysctl {CTL_KERN,KERN_PROC,KERN_PROC_PID,getpid()}; test kp_proc.p_flag & 0x800       | Confirmed |
| Process enumeration    | sub_9d29c                                         | sysctl KERN_PROC_ALL; size/0x288 = proc count; malloc; re-query full table            | Confirmed |
| Injected dylib (name)  | sub_16f54c / sub_16faf0                           | loop_dyld_image_count; _dyld_get_image_name; basename; strlen/compare                 | Confirmed |
| Injected code (region) | sub_2034ec / sub_1f32b4                           | mach_vm_region_recurse + proc_regionfilename + dladdr => region backed by file?       | Confirmed |
| Substrate / jailbreak  | sub_13a724 / sub_13a42c / sub_9865c               | access/statfs('/'), lstat/stat on target paths; decrypted cy:com.saurik.substrate     | High      |
| ObjC method swizzle    | sub_1eb5a0 / sub_1eb758                           | class_getInstanceMethod/ClassMethod; compare/observe IMP via method_setImplementation | High      |
| Inline hook / opcode   | (rule-driven)                                     | raises evt 0x0a set_inline_hook_error, 0x0b inline_hook_opcode_mismatch               | Confirmed |
| Timing check           | sub_d28f0 / sub_7e830                             | mach_absolute_time + mach_timebase_info deltas                                        | Medium    |
| Proxy / network env    | (ObjC) isProxy / CFNetworkCopySystemProxySettings | reads system proxy settings                                                           | Medium    |
| Screen capture         | UIScreenCapturedDidChangeNotification             | observes iOS screen-recording state                                                   | Medium    |

### Evidence — debugger self-check (sub\_37004)

```

37024: mov    w19,#0x288          ; sizeof(struct kinfo_proc)
37028: add    x0,sp,#0x20 ; bl _bzero ; zero result buffer
3703c: ldr    d0,[x8,#0x6c8]      ; mib[0..1]={CTL_KERN=1,KERN_PROC=14}
37044: mov    w8,#1 ; str w8,[sp,#0x18] ; mib[2]=KERN_PROC_PID=1
3704c: bl     _getpid ; str w0,[sp,#0x1c]; mib[3]=getpid()
37060: mov    w1,#4              ; namelen=4
3706c: bl     _sysctl              ; query own proc; caller tests P_TRACED (0x800)

```

Confidence: Confirmed. Canonical iOS anti-debugger pattern (KERN\_PROC\_PID on self).

## 5 Integrity System

Integrity is enforced on both *data* (downloaded rule/module blobs) and *code* (in-memory text). Failures map to concrete MRPCS events:

| Check               | Location / event                                           | Meaning                                                                   | Conf.     |
|---------------------|------------------------------------------------------------|---------------------------------------------------------------------------|-----------|
| Data CRC            | evt 0x00 ms_data_crc → mrpcs_data_crc_error                | CRC of a data blob mismatches expected                                    | Confirmed |
| Data length         | evt 0x01 ms_data_len; 0x02 ms_data_mod_len                 | blob / module-name length checks                                          | Confirmed |
| CRC report line     | sub_156f8                                                  | format "!!%s, size:%d, crc:%08x, t:%s" for reporting                      | High      |
| Hash notify         | sub_b0bfc                                                  | "notify mrpcs, hash:0x%08x"; mrpcs_abort.dat                              | High      |
| Inline-hook install | evt 0x0a set_inline_hook_error "%s;pointID:%s"             | expected hook-point could not be validated                                | Confirmed |
| Opcode mismatch     | evt 0x0b inline_hook_opcode_mismatch                       | bytes at a monitored code point != expected opcode => tamper              | Confirmed |
| Signature match     | evt 0x14 single_data_not_match; 0x15 common_data_not_match | memory-signature scan mismatch                                            | Confirmed |
| Code mirror / remap | sub_1488b8 / sub_2066bc                                    | vm_copy + vm_remap + mprotect + sys_icode_invalidate (clean-copy compare) | Medium    |

The **opcode-mismatch** event (0x0b) is the core code-integrity signal: the runtime keeps expected opcodes for monitored “point IDs” and raises a tamper event when the live bytes differ — the mechanism used to catch inline hooks. The expected opcodes and point IDs are supplied by the **rule** data (Section 7), not hard-coded, which is why the binary itself contains few literal signatures.

## 6 Anti-Instrumentation

Consolidated view of the environment/instrumentation defences and the exact APIs that evidence each. The debugger reporter sub\_f73dc builds "debugger :%s"/"debugger=%s" lines, tying the anti-debug detectors to the report channel.

| Surface                  | Evidence (APIs / offsets)                                               | Linked reporting               |
|--------------------------|-------------------------------------------------------------------------|--------------------------------|
| Debugger attach          | sysctl P_TRACED (sub_37004); syscall(sub_21d880/214de8)                 | debugger:%s reporter sub_f73dc |
| Task-port / pid_for_task | task_get_special_port sub_12824c; pid_for_task sub_2034ec/1f32b4        | foreign task inspection        |
| Injected dylibs          | _dyld_get_image_name loop sub_16f54c; dlopen/dlsym probes sub_1eef00    | module-name screening          |
| Injected/anon code       | mach_vm_region_recurse + proc_regionfilename sub_2034ec                 | unbacked executable regions    |
| Frida/Substrate          | decrypted cy:com.saurik.substrate; dlsym symbol probes sub_239bc0 (11x) | known-tool artefacts           |
| Jailbreak files          | access/stat/lstat/statfs sub_13a724/13a42c/f8030                        | path-existence checks          |
| ObjC hooking             | method IMP compare sub_1eb5a0; method_setImplementation sub_1eb758      | swizzle detection/monitoring   |
| Env vars                 | getenv (jailbreak/DYLD_* screening)                                     | process environment            |

**String obfuscation.** Sensitive artefacts are stored XOR-0x18 encrypted. Recovered plaintext includes cy:com.saurik.substrate, libdyld.dylib, and the update URL http://down.qq.com/iedsafe/Client/ios. Only a handful of detection artefacts are hard-coded — most target lists arrive as encrypted **rules** at runtime.

### Evidence — XOR-0x18 string decryption

```

cipher @0x26e9a8: 70 6c 6c 68 22 37 37 7c 77 6f 76 36 ... ^0x18 =>
'http://down.qq.com/iedsafe/Client/ios'
cipher @0x26e885 ^0x18 => 'cy:com.saurik.substrate' (MobileSubstrate id)
cipher @0x26e978 ^0x18 => 'libdyld.dylib'

```

*Confidence: Confirmed. Single-byte XOR key 0x18 recovered by known-plaintext (URL) and verified across strings.*



## 7 MRPCS Subsystem

MRPCS is the vendor's internal message/event core (platform tag MRPCS\_IOS). It owns the event vocabulary, the worker threads, the rule engine and the data-integrity checks. Its pieces:

| Role                | Offset     | Size   | Description                                                                  | Conf.     |
|---------------------|------------|--------|------------------------------------------------------------------------------|-----------|
| get_event_id        | sub_11ac6c | 0x18   | returns integer event id from an event object (accessor; 3 callers)          | High      |
| classify + emit     | sub_102d48 | 0x101c | 79-case switch: id -> (tag,desc); per-id config gate + once-flag; calls emit | Confirmed |
| emit / format       | sub_102b84 | 0x1a0  | snprintf "name=%s desc=%s"; hands string to router sub_234b4                 | Confirmed |
| command router      | sub_234b4  | 0x12bc | AnoSDKIoctlOld handler; 23 callers, 127 callees; central sink                | High      |
| hash notify / abort | sub_b0bfc  | -      | "notify mrpcs, hash:0x%08x"; mrpcs_abort.dat persistence                     | High      |
| platform tag        | sub_149f38 | -      | emits MRPCS_IOS identifier                                                   | Medium    |
| module loader       | sub_dccc0  | -      | "mrpcs_lib" module name handling                                             | Medium    |

**Worker threads.** MRPCS runs up to four workers; each start-failure has its own event (id 0x0f-0x12): f c thread (0x0f), scan thread (0x10), send thread (0x11), download thread (0x12). The **scan** thread runs memory/integrity checks; the **download** thread fetches rule blobs; the **send** thread flushes reports. (Section 11.)

**Rule engine.** Events 0x45 rule\_exe\_fail ("%s;pID:%s;rID:%s") and 0x4d rule\_op\_is\_change show MRPCS executes downloaded *rules* identified by point-ID / rule-ID. rule\_op\_is\_change firing means a rule's expected opcode changed in memory — i.e. the rule engine is itself a tamper tripwire. This is why detection logic is data-driven and updatable server-side without shipping a new binary.

## 8 Event System — IDs & Dispatching

The complete event catalogue below was decoded directly from the jump table at 0x2656fc (79 signed 32-bit offsets) inside sub\_102d48 — not from strings. For each live case the analyzer extracted the config-level *gate* (emit only if event . Level > N), the short *tag* and the *description* passed to the emit function. Empty slots share the default merge (reserved / unused ids).

| ID   | Case @   | Gate | Short tag                   | Description string                       |
|------|----------|------|-----------------------------|------------------------------------------|
| 0x00 | 0x102e0c | >5   | ms_data_crc                 | mrpcs_data_crc_error                     |
| 0x01 | 0x102e5c | >5   | ms_data_len                 | mrpcs_data_len_error                     |
| 0x02 | 0x102eac | >5   | ms_data_mod_len             | mrpcs_data_mode_name_len_error           |
| 0x03 | 0x102efc | >5   | ms_data_mode_infe_error     | %s;mod nm:%s                             |
| 0x06 | 0x102fe8 | >4   | fp:                         | %s;nm:%s err:%s                          |
| 0x08 | 0x1030f0 | >5   | get_mode_path_error         | %s;nm:%s                                 |
| 0x0a | 0x1031d8 | >5   | set_inline_hook_error       | %s;pointID:%s                            |
| 0x0b | 0x1032bc | >5   | inline_hook_opcode_dismatch | %s;pointID:%s                            |
| 0x0e | 0x1033a8 | >4   | download_data_failed        | %s;crc:%s                                |
| 0x0f | 0x10348c | >2   | ms_fc_start                 | fc_thread_statrt_failed!                 |
| 0x10 | 0x1034dc | >2   | ms_scan_start               | mrpcs_scan_thread_start_failed!          |
| 0x11 | 0x10352c | >2   | ms_send_start               | mrpcs_send_data_thread_start_failed!     |
| 0x12 | 0x10357c | >5   | ms_down_start               | mrpcs_download_data_thread_start_failed! |
| 0x14 | 0x1035d0 | >5   | ms_single_match             | mrpcs_single_data_not_match!             |
| 0x15 | 0x103620 | >5   | ms_common_match             | mrpcs_common_data_not_match!             |
| 0x19 | 0x10367c | >3   | send_size_beyond_buff       | %s;o_sz:%s; bf_sz:%s; pID:%s             |
| 0x2c | 0x1037ec | >5   | mmap_fialed                 | %s;error:%s                              |
| 0x38 | 0x1038fc | >3   | ms_push_game                | send_data_push_game_data_failed          |
| 0x39 | 0x10394c | >3   | ms_game_remain              | send_data_to_game_data_remain_count      |
| 0x3a | 0x10399c | >3   | ms_game_data_toolong        | send_data_to_game_pack_data_too_long     |
| 0x3b | 0x1039ec | >1   | not_find_mode_base          | %s;mod_nm:%s                             |
| 0x45 | 0x103af4 | >1   | rule_exe_fail               | %s;pID:%s;rID:%s                         |
| 0x4d | 0x103c14 | >1   | rule_op_is_change           | %s;pID:%s;rID:%s                         |

### Who does what (the questions the analysis set out to answer):

| Stage                  | Owner (evidenced)                                                                  |
|------------------------|------------------------------------------------------------------------------------|
| Detect the condition   | the collectors/detectors (Sec.4-6) and rule engine set the trigger                 |
| Assign the Event ID    | sub_11ac6c get_event_id(event) -> integer 0..78                                    |
| Interpret the Event ID | sub_102d48 switch (jump table 0x2656fc) selects the case                           |
| Gate / dedup           | per-id config-level compare + per-id once-flag byte at 0x297fb1+ (cleared on emit) |
| Build the description  | sub_102d48 loads (tag,desc) literals; sub_102b84 formats name=.. desc=..           |
| Record / route         | sub_234b4 (command router) ingests the formatted event                             |
| Send                   | report queue (Sec.9) + senders (Sec.10)                                            |

## 9 Reporting Pipeline

From a raised condition to a stored report:

condition → get\_event\_id (sub\_11ac6c) → classify (sub\_102d48, gate+once-flag) → emit/format (sub\_102b84, "name=<tag>|desc=<description>") → command router (sub\_234b4) → report queue.

| Stage           | Offset             | Role                                                            | Conf.     |
|-----------------|--------------------|-----------------------------------------------------------------|-----------|
| emit/format     | sub_102b84         | builds name= desc=; calls sub_234b4                             | Confirmed |
| router / sink   | sub_234b4          | AnoSDKIoctlOld handler; ingests events & commands (127 callees) | High      |
| queue: read     | sub_1a25c          | AnoSDKGetReportData — host polls pending report buffer          | High      |
| queue: delete   | sub_1a0cc          | AnoSDKDelReportData — host acks/removes a delivered report      | High      |
| queue: recv     | sub_1a278          | AnoSDKOnRecvData — server->client (delivers new rules)          | High      |
| variant getters | GetReportData2/3/4 | sub_24dc0 / sub_2dc90 / sub_2dd5c — alternate report formats    | Medium    |

### Evidence — emit formats the event record (sub\_102b84)

```

102b84: ... sprintf(buf,"name=%s", tag)
        ... sprintf(buf2,"|desc=%s", desc)
        ... sprintf(out,"%s%s", buf, buf2)    ; 'name=<tag>|desc=<desc>'
        bl 0x234b4                          ; -> command router / report sink

```

Confidence: Confirmed. Strings 'name=%s','|desc=%s','%s%s' are the only literals in sub\_102b84; single callee sub\_234b4.

## 10 Network / Send Pipeline

Two egress paths were confirmed:

**Path A — host callback (primary).** The host game registers `tss_sdk_send_data_to_svr` (wired in `sub_f0590`, log line `"init_info->tss_sdk_send_data_to_svr:%p"`). Reports queued for `AnoSDKGetReportData` are pulled by the game and sent over the game's own channel; the game returns server data via `AnoSDKOnRecvData`.

**Path B — native socket (direct).** A self-contained TCP client: `sub_f8148` connect, `sub_f8210` send, `sub_f829c` recv, with `getaddrinfo` (`sub_7d870`), `setsockopt` (`sub_5a9f8`) and `select/poll` readiness. The higher-level buffer `sub_5a654` (11 callers) feeds send.

| Stage                | Offset                       | Evidence                                                               | Conf.     |
|----------------------|------------------------------|------------------------------------------------------------------------|-----------|
| send-callback wiring | <code>sub_f0590</code>       | stores host <code>tss_sdk_send_data_to_svr</code> fn-ptr               | Confirmed |
| socket connect       | <code>sub_f8148</code>       | <code>_connect</code> (caller <code>sub_7904</code> )                  | Confirmed |
| socket send          | <code>sub_f8210</code>       | <code>_send</code> (caller <code>sub_5a654</code> buffer, 11 callers)  | Confirmed |
| socket recv          | <code>sub_f829c</code>       | <code>_recv</code> (4 callers incl. poll loop <code>sub_b2eb8</code> ) | Confirmed |
| DNS resolve          | <code>sub_7d870</code>       | <code>_getaddrinfo</code>                                              | High      |
| sockopt              | <code>sub_5a9f8/5aa94</code> | <code>_setsockopt</code> (timeouts / keepalive)                        | High      |
| send stats           | <code>sub_9a968</code>       | <code>"send_cs:%d, send_gs:%d, remain_cnt:%d"</code>                   | Medium    |

*Send-pipeline faults are themselves events: `0x11 send-thread start fail`, `0x19 send_size_beyond_buff`, `0x38 push_game_data_failed`, `0x39 game_data_remain`, `0x3a game_data_toolong` — i.e. the transport is self-monitored.*

## 11 Threads

Five `pthread_create` sites were located. Four correspond to the MRPCS workers named by their start-failure events (0x0f-0x12); one entry point (`sub_189338`) resolves directly, the others take their entry from a worker/thread-pool struct ("ThreadPool3"), so entry recovery there is indirect.

| Thread                | Creator (site)             | Entry                                             | Purpose (inferred)                                            | Linked event                          | Conf.  |
|-----------------------|----------------------------|---------------------------------------------------|---------------------------------------------------------------|---------------------------------------|--------|
| MRPCS scan worker     | sub_1894a4 (site 0x189550) | indirect                                          | memory / integrity / signature scan loop                      | evt 0x10 scan_thread_start_failed     | High   |
| MRPCS send worker     | sub_1b94b4 (site 0x1b950c) | indirect                                          | flush report queue to sender                                  | evt 0x11 send_thread_start_failed     | High   |
| MRPCS download worker | sub_1ef610 (site 0x1ef6d4) | indirect (uses <code>pthread_threadid_np</code> ) | fetch rule/module blobs from server                           | evt 0x12 download_thread_start_failed | High   |
| fc worker             | sub_189294 (site 0x1892f4) | sub_189338 (direct)                               | periodic frequency/heartbeat check task                       | evt 0x0f fc_thread_start_failed       | Medium |
| named worker / pool   | sub_3e368 (site 0x3e424)   | indirect (+ <code>pthread_setname_np</code> )     | general worker; sets thread name; stacksize/detach configured | -                                     | Medium |

Shared state is guarded by a full `pthread_mutex` / `pthread_once` suite (imports present) and a `dispatch_semaphore`; the report queue (Sec.9) is the primary cross-thread structure (producers = scan/detector threads, consumer = send thread + host `GetReportData`).

### Evidence — thread creation site (sub\_189294)

```

189294: ... pthread_attr_init / pthread_attr_setdetachstate
1892e0: adrp/add x2, 0x189338          ; thread entry pointer (sub_189338)
1892f4: bl _pthread_create             ; start worker
      (start-failure path emits MRPCS fc/scan/send/download event)

```

*Confidence: Medium-High. Direct entry only for sub\_189338; other workers load entry from a pool struct (indirect).*

## 12 Function Database

Consolidated table of the anti-cheat-relevant routines recovered. Names beginning `_AnoSDK / PluginTss` are real symbols; the rest are analyst proposals over stripped `sub_<offset>` functions. “Ret” summarises the observed return convention.

| Offset   | Proposed name                      | Role                                         | Callers                  | Callees                      | Ret      | Conf.     |
|----------|------------------------------------|----------------------------------------------|--------------------------|------------------------------|----------|-----------|
| 0xf0fd0  | <code>_AnoSDKInit</code>           | host init entry                              | host                     | <code>sub_2d69c</code>       | W0 bool  | Confirmed |
| 0x2d69c  | <code>ac_init_internal</code>      | real initialization                          | <code>AnoSDKInit</code>  | <code>sub_25ad8,6bbfc</code> | status   | High      |
| 0x25ad8  | <code>ac_bootstrap</code>          | subsystem bootstrap (getpid/kill guard)      | <code>sub_2d69c</code>   | 11 subsys                    | -        | Medium    |
| 0x6bbfc  | <code>get_context</code>           | pthread_once global context                  | 204 callers              | <code>sub_7f320..</code>     | ptr X0   | High      |
| 0x18068  | <code>get_manager</code>           | global manager accessor                      | 170 callers              | 1 callee                     | ptr X0   | Medium    |
| 0xf0490  | <code>log_format</code>            | vsnprintf log/format helper                  | 73 callers               | <code>vsprintf</code>        | -        | High      |
| 0x11ac6c | <code>get_event_id</code>          | event object -> id                           | 3 callers                | (leaf)                       | W0 int   | High      |
| 0x102d48 | <code>event_classify_emit</code>   | 79-case id->(tag,desc) + gate                | <code>sub_11af80</code>  | <code>sub_102b84</code>      | bool X0  | Confirmed |
| 0x102b84 | <code>emit_event</code>            | format name=.. desc=..                       | <code>sub_102d48</code>  | <code>sub_234b4</code>       | void(SE) | Confirmed |
| 0x234b4  | <code>cmd_report_router</code>     | <code>AnoSDKIoctlOld / event+cmd sink</code> | 23 callers               | 127                          | status   | High      |
| 0x26398  | <code>ioctl_router_new</code>      | <code>AnoSDKIoctl</code> handler             | <code>AnoSDKIoctl</code> | -                            | status   | High      |
| 0x11af80 | <code>event_dispatch_loop</code>   | iterate & dispatch events (blr vtable)       | (indirect)               | <code>sub_102d48+</code>     | void(SE) | High      |
| 0x1a25c  | <code>report_queue_get</code>      | <code>AnoSDKGetReportData</code>             | 2 callers                | 1                            | ptr/len  | High      |
| 0x1a0cc  | <code>report_queue_del</code>      | <code>AnoSDKDelReportData</code>             | 2 callers                | 2                            | void(SE) | High      |
| 0x1a278  | <code>on_rcv_data</code>           | <code>AnoSDKOnRcvData</code> (rules in)      | 2 callers                | 2                            | status   | High      |
| 0x37004  | <code>chk_debugger</code>          | P_TRACED self-check                          | (detector)               | <code>sysctl</code>          | W0 bool  | Confirmed |
| 0x9d29c  | <code>enum_processes</code>        | KERN_PROC_ALL process list                   | (collector)              | <code>sysctl,malloc</code>   | count    | Confirmed |
| 0x16f54c | <code>scan_dyld_images</code>      | loop loaded image names                      | (detector)               | <code>sub_16faf0</code>      | W0 flag  | Confirmed |
| 0x16faf0 | <code>get_image_name</code>        | dyld image name+basename                     | <code>sub_16f54c</code>  | <code>dyld_*</code>          | ptr/len  | Confirmed |
| 0x2034ec | <code>scan_vm_regions</code>       | region->file (injected code)                 | <code>sub_203dac</code>  | <code>vm_region+proc</code>  | W0 flag  | Confirmed |
| 0x13a724 | <code>chk_path_access</code>       | access/statfs jailbreak path                 | (detector)               | <code>access,statfs</code>   | W0 bool  | High      |
| 0x13a42c | <code>chk_path_stat</code>         | lstat/stat jailbreak path                    | (detector)               | <code>lstat,stat</code>      | W0 bool  | High      |
| 0x1eb5a0 | <code>chk_objc_imp</code>          | compare ObjC method IMP                      | (detector)               | <code>class_get*</code>      | W0 bool  | High      |
| 0x33d6c  | <code>device_id_store</code>       | keychain device fingerprint                  | (collector)              | <code>SecItem*</code>        | ptr      | High      |
| 0xf73dc  | <code>report_debugger</code>       | build debugger:%s report                     | (reporter)               | -                            | void(SE) | High      |
| 0x156f8  | <code>crc_report_fmt</code>        | format crc:%08x report line                  | <code>sub_14fb0</code>   | <code>sub_17518</code>       | void(SE) | Medium    |
| 0xb0bfc  | <code>mrpcs_notify</code>          | notify mrpcs hash / abort.dat                | (mrpcs)                  | -                            | void(SE) | High      |
| 0xf0590  | <code>set_send_cb</code>           | wire tss_sdk_send_data_to_svr                | 1 caller                 | <code>sub_f0490</code>       | void(SE) | High      |
| 0xf8210  | <code>socket_send</code>           | native_send                                  | <code>sub_5a654</code>   | <code>_send</code>           | W0 n     | Confirmed |
| 0xf829c  | <code>socket_rcv</code>            | native_rcv                                   | 4 callers                | <code>_recv</code>           | W0 n     | Confirmed |
| 0xf8148  | <code>socket_connect</code>        | native_connect                               | <code>sub_7904</code>    | <code>_connect</code>        | W0       | Confirmed |
| 0x28da70 | <code>PluginTssSDKLifecycle</code> | ObjC app-lifecycle bridge                    | runtime                  | <code>objc</code>            | obj      | Confirmed |



## 13 Critical Functions

### **sub\_102d48 — event\_classify\_emit**

The hinge of the whole system. Its 79-entry jump table *is* the event vocabulary; everything a detector finds becomes meaningful only after passing through here. Owns per-event gating and the once-flags. Removing/altering it would blind reporting entirely — which is exactly why it is the highest-value function to understand and protect.

### **sub\_234b4 — cmd\_report\_router**

AnoSDKIoctlOld: 23 callers, 127 callees. The convergence point where classified events and host ioctl commands meet the report machinery. The single busiest routing function.

### **sub\_6bbfc / sub\_18068 — global context/manager**

204 / 170 callers. The shared-state singletons every subsystem reaches through; the natural anchors for mapping data structures.

### **sub\_11af80 — event\_dispatch\_loop**

Drives classification via indirect (blr) dispatch — the vtable-style indirection that makes static call-graphing incomplete; worth dynamic confirmation.

### **sub\_37004 / sub\_9d29c / sub\_2034ec — the confirmed detectors**

The three highest-confidence detection primitives (debugger, process list, injected-region scan); each is a self-contained, individually testable check.

### **sub\_f0590 + sub\_f8210 — the egress**

Where results leave the process. The two send paths (host callback + native socket) define the network surface of the anti-cheat.



## 14 Full Call Graph (anti-cheat core)

Condensed static call graph of the confirmed core. [SE] = side-effect / return not consumed by caller; (blr) = indirect dispatch (target resolved via register/vtable at runtime).

```
AnoSDKInit (0xf0fd0)
|__ ac_init_internal sub_2d69c
|__ ac_bootstrap sub_25ad8 (getpid+kill guard; spawns MRPCS workers)
|__ get_context sub_6bbfc [204 callers, singleton]
|__ get_manager sub_18068 [170 callers, singleton]

[ObjC] PluginTssSDKLifecycle / CloudAppLifecycleObserver
|__ applicationDidBecomeActive: -> AnoSDKOnResume (sub_2dc60, code=3)
|__ applicationDidEnterBackground:-> AnoSDKOnPause (sub_2dc60, code=2)
|__ DoCmdOnMainThread: / SendCmd: -> command router sub_234b4

MRPCS worker threads (pthread_create)
|__ scan sub_1894a4 (blr entry) -> detectors + integrity
|__ send sub_1b94b4 (blr entry) -> socket_send sub_f8210
|__ download sub_1ef610 (blr entry) -> on_recv_data / rule install
|__ fc sub_189294 -> sub_189338

Detection / Integrity (each raises a condition)
|__ chk_debugger sub_37004 (sysctl P_TRACED) -> event
|__ enum_processes sub_9d29c (sysctl KERN_PROC_ALL)
|__ scan_dyld_images sub_16f54c -> get_image_name sub_16faf0
|__ scan_vm_regions sub_2034ec (vm_region_recurse+proc_regionfilename)
|__ chk_path_access sub_13a724 / chk_path_stat sub_13a42c (jailbreak)
|__ chk_objc_imp sub_1eb5a0 (method IMP compare)
|__ crc/opcode integrity -> events 0x00 / 0x0a / 0x0b

MRPCS event core
event_dispatch_loop sub_11af80
|__ get_event_id sub_11ac6c -> W0 = id (0..78)
|__ event_classify_emit sub_102d48 (jump table 0x2656fc)
|   |__ gate: level compare + once-flag @0x297fb1+
|   |__ emit_event sub_102b84 [SE] ("name=<tag>|desc=<desc>")
|   |__ cmd_report_router sub_234b4 [23 callers/127 callees]
|__ (blr x9/x10) per-field handlers sub_11a7a0 .. sub_11a9cc

Report queue (host ABI)
|__ report_queue_get sub_1a25c (AnoSDKGetReportData) --> host
|__ report_queue_del sub_1a0cc (AnoSDKDelReportData)
|__ on_recv_data sub_1a278 (AnoSDKOnRecvData) <-- server rules

Senders
|__ Path A: set_send_cb sub_f0590 -> host tss_sdk_send_data_to_svr
|__ Path B: socket_connect sub_f8148 -> socket_send sub_f8210 -> socket_recv sub_f829c
```

## 15 Event-Flow Examples

Two concrete end-to-end traces, each annotated with the real offsets.

### Example A — inline-hook / opcode tamper (event 0x0b)

```
condition : a monitored code point's live bytes != expected opcode (rule-supplied)
|
detector  : integrity/rule check (scan worker sub_1894a4) sets the trigger for id 0x0b
|
event id   : get_event_id sub_11ac6c -> 0x0b
|
classify  : sub_102d48 case 0x0b @0x1032bc
            gate   : event.level > 5 ?           (config threshold)
            once   : flag @0x297fbc set & clear (report-once)
            (tag,desc) = ("inline_hook_opcode_mismatch","%s;pointID:%s")
|
emit      : sub_102b84 -> "name=inline_hook_opcode_mismatch|desc=<pID.>"
|
route     : cmd_report_router sub_234b4
|
queue     : report_queue_get sub_1a25c (host polls) / or
send      : socket_send sub_f8210
```

### Example B — debugger attached (P\_TRACED)

```
condition : sysctl(KERN_PROC_PID,self).kp_proc.p_flag & P_TRACED (0x800)
|
detector  : chk_debugger sub_37004 -> W0 = traced?
|
report    : report_debugger sub_f73dc builds "debugger:%s" / "debugger=%s"
|
route     : cmd_report_router sub_234b4 (as an MRPCS record)
|
egress    : Path A host callback tss_sdk_send_data_to_svr (sub_f0590) OR
            Path B native socket_send sub_f8210
```

## 16 Evidence Appendix

Primary disassembly excerpts underpinning the key claims. All offsets are file/VA (identical for \_\_TEXT here).

### A16.1 — MRPCS event jump table (sub\_102d48)

#### switch selection over 79 ids

```
102de4: ldr w8,[sp,#0x168]          ; w8 = event id (from get_event_id)
102de8: cmp x8,#0x4e                ; 78 = max id
102df0: b.hi 0x103d1c               ; default
102df4: adrp x8,0x265000; add x8,#0x6fc; = 0x2656fc (jump table base)
102e00: ldrsw x10,[x8, x11, lsl #2] ; signed 32-bit offset[id]
102e04: add x9,x8,x10 ; br x9       ; -> case handler
-- case 0 (id 0x00) @0x102e0c --
102e10: ldr w9,[x8] ; cmp w9,#5 ; b.le skip ; config-level gate
102e24: ldrb w8,[0x297fb1] ; tbz -> proceed ; once-flag
102e48: add x1,#'ms_data_crc' ; x2,#'mrpcs_data_crc_error'
102e54: bl 0x102b84                ; emit_event(ctx,tag,desc)
```

Confidence: Confirmed. 79 entries read from 0x2656fc; 23 live cases decoded (Section 8).

### A16.2 — process enumeration vs. self-debug check

#### two distinct sysctl MIBs

```
sub_9d29c: MIB @0x24c2d0 = {1,14,0,0} = {CTL_KERN,KERN_PROC,KERN_PROC_ALL,0}
          size/0x288 (sizeof kinfo_proc) => process count; malloc; re-query
sub_37004: MIB = {1,14,1,getpid()} = {CTL_KERN,KERN_PROC,KERN_PROC_PID,pid}
          buffer 0x288; caller tests kp_proc.p_flag & 0x800 (P_TRACED)
```

Confidence: Confirmed. KERN\_PROC\_ALL = enumerate all; KERN\_PROC\_PID(self) = anti-debug.

### A16.3 — injected image / region scanning

#### module-name loop + region->file

```
sub_16faf0: _dyld_get_image_header(i); _dyld_get_image_name(i);
          find last '/' (sub_16f960 basename); strlen; compare (<0x64)
sub_2034ec: _pid_for_task; mach_vm_region_recurse(...);
          proc_regionfilename(pid,addr,...); dladdr ; region backed by file?
```

Confidence: Confirmed. Anonymous executable regions / unexpected image names = injection signals.

### A16.4 — XOR-0x18 string obfuscation

#### recovered plaintext

```
@0x26e9a8 ^0x18 = 'http://down.qq.com/iedsafe/Client/ios' (Tencent update)
@0x26e885 ^0x18 = 'cy.com.saurik.substrate' (MobileSubstrate)
@0x26e978 ^0x18 = 'libdyld.dylib'
```

Confidence: Confirmed. Key 0x18 recovered by known-plaintext; consistent across all encrypted strings tested.

### A16.5 — emit / router linkage

#### event record format and sink

```
sub_102b84 literals: 'name=%s' , '|desc=%s' , '%s%s' ; single callee sub_234b4
_AnoSDKIoctl0ld (0xf1168) tail-calls sub_234b4 ; router = ioctl handler
sub_234b4: 23 callers, 127 callees ; central report/command sink
```

Confidence: High. emit\_event feeds the same router the host ioctl uses.

### A16.6 — method & scope note

Analysis was performed statically on Linux using llvm-objdump (Mach-O ARM64 disassembly with stub→symbol and literal-pool annotations), lief (Mach-O parsing, sections, symbols, \_\_mod\_init\_func), and a custom Python engine that recovered 10,551 function boundaries from BL/branch targets + constructors + code-pointer references, built the direct/indirect call graph, and resolved ADRP/ADD literal xrefs — the same artefacts a Ghidra auto-analysis + decompiler + reference view would produce. Indirect blr targets

and rule-driven detection thresholds cannot be fully resolved statically and are marked Medium/Unknown where relevant; dynamic instrumentation was **not** used. No patching, bypass, hook, or detection-defeating modification was performed or is described anywhere in this document — the deliverable is a structural map for defensive auditing only.

## ★ Anti-Cheat Master Map

The whole runtime on one page — real offsets substituted for generic stage names.

```

HOST GAME (links anogs.framework)
  | AnoSDKInit 0xf0fd0 / SetUserInfo 0xf1000 / OnPause|OnResume 0xf109c/0xf10bc
  v
ENTRY / INIT ..... sub_2d69c -> ac_bootstrap sub_25ad8
  | singletons: get_context sub_6bbfc [204] , get_manager sub_18068 [170]
  | objc bridge: PluginTssSDKLifecycle (app foreground/background)
  v
COLLECTORS ..... device-id sub_33d6c (keychain) | process-list sub_9d29c
  | dyld-image-list sub_16f54c | vm-region-map sub_2034ec
  v
DETECTORS ..... debugger sub_37004 (P_TRACED) | objc-swizzle sub_1eb5a0
  | jailbreak/substrate sub_13a724/13a42c | dlsym-probe sub_239bc0
  v
INTEGRITY ..... data-CRC (evt 0x00) | opcode-mismatch (evt 0x0b)
  | signature-match (evt 0x14/0x15) | code-mirror sub_1488b8
  v
CLASSIFIER ..... get_event_id sub_11ac6c -> event_classify_emit sub_102d48
  | jump table 0x2656fc (79 ids) + gate + once-flags 0x297fb1+
  v
EVENTS ..... 23 live ids: crc / len / hook / opcode / match / thread /
  | mmap / send / rule_exe_fail / rule_op_is_change (Sec.8)
  v
DISPATCH ..... event_dispatch_loop sub_11af80 (blr vtable dispatch)
  v
REPORT_BUILDER ..... emit_event sub_102b84 "name=<tag>|desc=<desc>"
  v
ROUTER / SINK ..... cmd_report_router sub_234b4 (AnoSDKIoctl0ld; 23<-/127->)
  v
QUEUE ..... GetReportData sub_1a25c | DelReportData sub_1a0cc
  | OnRecvData sub_1a278 <== server rules (download worker sub_1ef610)
  v
SENDERS ..... Path A: tss_sdk_send_data_to_svr (wired sub_f0590) --> host
  | Path B: connect sub_f8148 -> send sub_f8210 -> recv sub_f829c
  | endpoint (decrypted): down.qq.com/iedsafe/Client/ios

```

*End of report. Defensive structural analysis only — read-only; no modification, bypass, or evasion technique is included.*